CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 11: **Electricity** | Quick revision, one page

Charge, current, potential difference

- Current $I = \dfrac{Q}{t}$ (unit ampere, A = C/s); charge of an electron $= 1.6 \times 10^{-19}$ C; 1 C $\approx 6 \times 10^{18}$ electrons.
- Potential difference $V = \dfrac{W}{Q}$ (unit volt, V = J/C). Conventional current flows from the positive to the negative terminal.
- Ammeter: in series, low resistance. Voltmeter: in parallel, high resistance.

Ohm's law and resistance

- $V = IR$ (at constant temperature); R in ohm (Ω).
- $R = \rho\dfrac{l}{A}$; ρ is resistivity (unit Ω m), depends on material and temperature, not on size. Resistance $\propto l$, $\propto \dfrac{1}{A}$.
- Metals: low ρ; alloys: higher ρ, do not oxidise readily (nichrome in heaters); tungsten filament for bulbs (high melting point).

Combinations

- Series: $R_s = R_1 + R_2 + R_3$; same current, voltage divides, $V = V_1 + V_2 + V_3$.
- Parallel: $\dfrac{1}{R_p} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dfrac{1}{R_3}$; same voltage, current divides, $I = I_1 + I_2 + I_3$; R_p is less than the smallest resistor.
- Two equal resistors R: series $2R$, parallel $\dfrac{R}{2}$.

Heating effect, power and energy

- $H = I^2Rt$ (joule's law of heating, H in J, t in s).
- Power $P = VI = I^2R = \dfrac{V^2}{R}$ (unit watt, W). Energy $E = Pt$.
- 1 kWh $= 3.6 \times 10^6$ J (one unit of electrical energy). Uses: electric heater, bulb filament, fuse (low melting point).

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